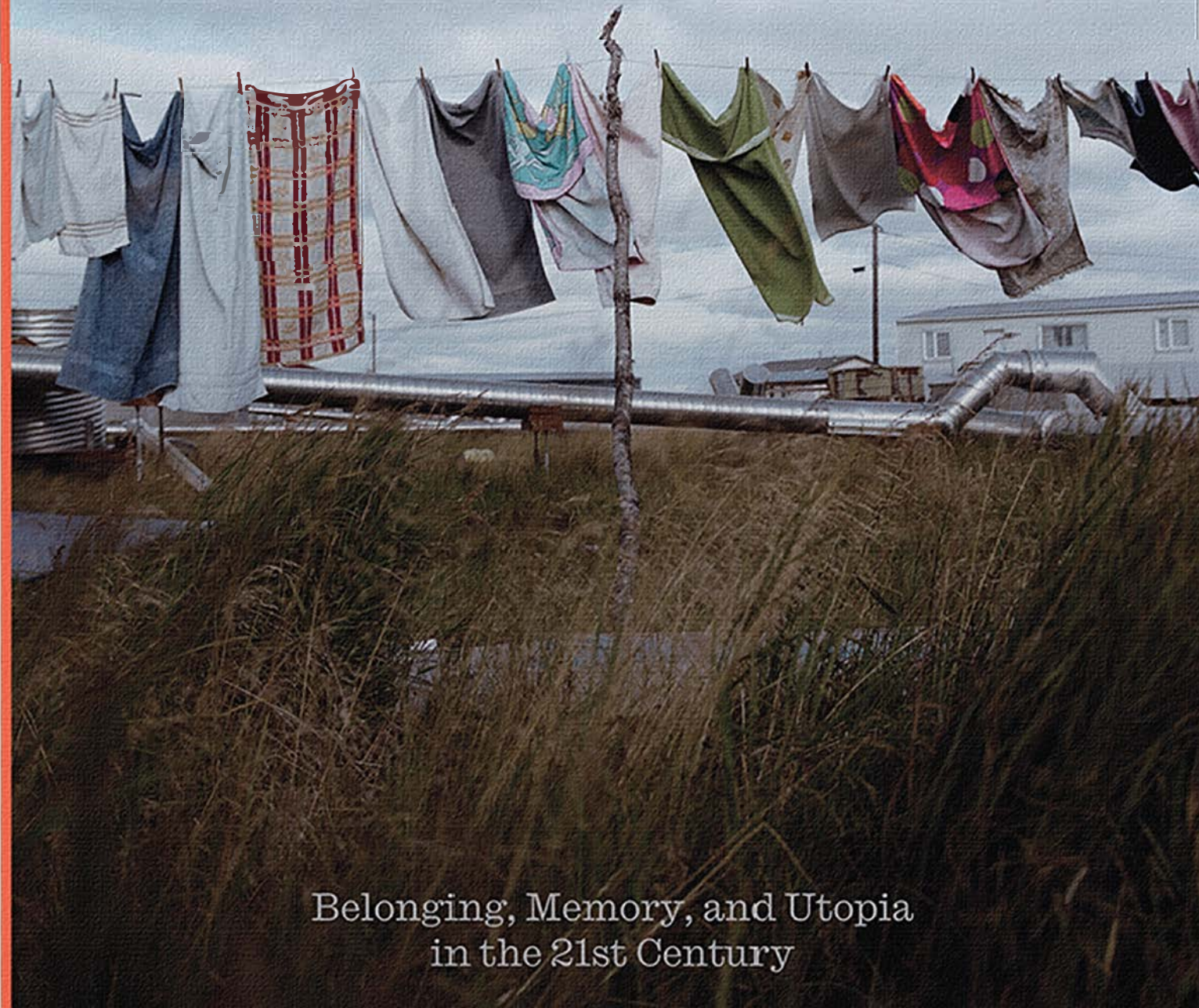


Smithsonian Design Triennial

MAKING HOME



Belonging, Memory, and Utopia
in the 21st Century

DRAW ME THE EARTH WHOLE

There will be unexpected consequences. Taking care of unexpected country will be required—again and always. Reconciliation is not guaranteed; it is proffered, suggested, haltingly pictured. Any reconciliation will depend on descendants of settler worlds letting go of salvation history.... Technocultural people must study how to live in actual places, cultivate practices of care, and risk ongoing face-to-face encounters with unexpected partners.

DONNA HARAWAY, *Speculative Fabulations for Technoculture's Generations*^[1]

We live in an epoch that is shaped by climate catastrophes with risks and uncertainties on a planetary scale. On graph after graph, in metric after metric—carbon dioxide in the atmosphere, species extinction, particulate matter in the air—the rate of climate change is referred to as the “great acceleration.” A large amount of scientific research and images have sought to communicate the destructive impacts of the climate crisis. And yet, the climate story is difficult both to tell and to hear, both to comprehend and to act on. For scientists, journalists, activists, and designers, the question of how to bring climate matters to public concern is open and urgent. How might the architectural imagination make sense of the Earth at a moment in which the planet is presented in crisis?

Climate, meaning the statistics of weather over a defined time span, does not manifest itself in any single moment, event, or location and can never be directly experienced. In contrast to the immediately experienceable weather event, climate is inherently mediated. The only way it can be apprehended is through systems of data and modeling, including cultural forms—narratives, vocabularies, images, objects, and myths. In other words, climate—as it is imagined and acted upon—needs to be understood culturally. “Climate,” writes geographer Mike Hulme, “is weather which has been cultured, interpreted, and acted on by the imagination, through storytelling and material technologies.”^[2] Approaching climate this way, adds Hulme, demands an “explicitly *geographical* and cultural interrogation of how people

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DESIGN EARTH, Blue Marble Circus, 2017

live climatically.”^[3] After all, “climate,” derived from the Greek *klinein* (to lean, rest, recline, bend), was originally a geographical term denoting a position on the Earth defined by latitude (i.e., the specific inclination of the sun on a given place at summer solstice).^[4] Retrieving this meaning of climate—geographically situated and differentiated—means considering climate not only from the standpoint of earth sciences

as a global statistical average, but also as a situated cultural project of the geographic imagination. Storytelling becomes a method by which to make sense of and relate to a here and now that is otherwise difficult to tell and hear and engenders ways of seeing and knowing the many externalities of our time.

For the collaborative studio practice DESIGN EARTH, storytelling has become a method to act on climate change by the geographic imagination, through geostories. In my book coauthored with El Hadi Jazairy, *Geostories: Another Architecture for the Environment*, we put forth an “architecture with externalities,” one that brings into representation—into aesthetics and politics—those things, spaces, and scales otherwise not depicted in the popular imagination of the planet. The series of speculative narratives imagines how technological systems have changed the Earth and considers ways of living *with* externalities—with landfills, abandoned mine pits, decommissioned oil rigs, and space debris. Here, the architectural narrative, as expounded through drawings, narratives, and artifacts, becomes a heuristic device that incorporates externalities to give both figurative shape and cultural meaning to matter and trouble whose repercussions are dispersed across space and time and whose actual costs often remain unaccounted for. These geostories are geographically situated speculations: they always depart from a measured here and a controversial now. Yet, they are not documentary. Nor are they completely fictional. Rather, representation becomes a way to work with uncertainties, frictions, and contradictions, all while

magnifying the cultural and geographic climates they engage.

Stories and images matter for the Earth. The present predicament of the climate crisis springs from narratives of modernity—about the world as resource, about nature as external, and about progress as an escape from the gravity of geography. ^[5] In this worldview, the Earth—from the

[1] Donna Haraway, “Speculative Fabulations for Technoculture’s Generations: Taking Care of Unexpected Country,” *Australian Humanities Review*, Issue 50, May 2011. <https://australianhumanitiesreview.org/2011/05/01/speculative-fabulations-for-technocultures-generations-taking-care-of-unexpected-country/>.

[2] Mike Hulme, “Chapter 1: What Is Climate?” *Weathered: Cultures of Climate* (London: SAGE Publications, 2017).

[3] *Ibid.*, 57.

[4] Eva Horn, “Air as Medium,” *Grey Room*, no. 73 (2018): 6–25.

[5] Christophe Bonneuil, “The Geological Turn,” in *The Anthropocene and the Global Environmental Crisis: Rethinking Modernity in a New Epoch*, ed. Clive Hamilton, François Gemenne, and Christophe Bonneuil (London: Routledge, 2015), 17.

atmosphere to the geological superstratum—is a resource predicated on the discounting of negative externalities and their erasure from the cultural imagination. Think of the vast extractive network of fossil fuel extraction, for example. The largest single commodity in international trade, both in value and in weight, oil has inscribed its planetary infrastructural order. Over the past two hundred years, and in particular since the Second World War, the economic promises of carbon modernity have rested on the industry's abstraction and denial of its hidden costs—of land degradation, population decimation, asthma and cancer cases, surface collapse, contamination of subsurface water flows, and the exponential increase in methane and carbon dioxide emissions. In economic discourse, the term "externalities" describes the negative social costs or ill effects of private calculation, notably pollution, public health, and the degradation of shared resources (such as air and water); instead, they are shifted to and ultimately borne by third parties, that is, "other" species, generations or temporalities, people or geographies. Violence, both physical and symbolic, is inherent to the geography of externalities. Acts of destruction—always racialized, classed, and gendered—are validated through stories and images that justify the selective devastation and erasure of "other" worlds and legitimize dispossession through worldviews such as Manifest Destiny, terra nullius, or orientalism.

New stories and images are necessary to participate in an ethical and just response to climate change. The climate crisis began to occupy public consciousness when the unwanted consequences of extractivist actions did not disappear from view. There was "no zone of reality in which we could casually rid ourselves of the consequence of human political, industrial, and economic life."^[6] The historical importance of ecological crises, French philosopher Bruno Latour pointed out, "stems from the impossibility of continuing to imagine politics on the one side and, on the other, a nature that would serve politics simultaneously as a standard, a foil, a reserve, a resource, and a public dumping ground."^[7] Latour adds: "Why not transform this whole business of recalling modernity into a grand question of design?"^[8] His response calls for crafting the "political arts," an experimental method that takes action on climate change by connecting political ecology with aesthetic experience. In this worldview, the role of representation shifts from reinforcing consensus on solutioneering to assembling publics around climate controversies. The reason for which this is needed, Latour states, is that "the Earth has become once again ... an agent of ... our common geostory." He continues, "The

[6] Bruno Latour, *Politics of Nature: How to Bring the Sciences into Democracy* (Cambridge, MA: Harvard University Press, 2004), 58.

[7] Ibid.

[8] Bruno Latour, *An Inquiry into Modes of Existence: An Anthropology of the Moderns* (Cambridge, MA: Harvard University Press, 2013), 23.

[9] Bruno Latour, "Agency at the Time of the Anthropocene," *New Literary History*, no. 45 (2014): 3.

problem becomes for all of us in philosophy, science, or literature, how do we tell such a story."^[9]

What would happen if new narratives revolved around things previously relegated as externalities? And which techniques of representation might help make them public? Perhaps speculative narrative and its persistent engagement with technological questions in relation to the planet, including those created by broadening risk scenarios, can open up earthly stories to the urgent troubles of externalities. In her practice of SF, a referent to "science fiction, speculative fabulation, string figures, speculative feminism, science fact," Donna Haraway reminds us that "it matters what matters we use to think other matters with; it matters what stories we tell to tell other stories with.... It matters what stories make worlds, what worlds make stories."^[10] In *Staying with the Trouble*, Haraway describes what she has come to call "speculative fabulation" as a "mode of attention, a theory of history, and a practice of worlding" for imagining future presents that disrupt habitual and binary ways of knowing—to open up Earth stories to urgent troubles.

Such narratives dwell within the fundamentally unhome-ly histories of environmental violence—imbued and perpetuated with colonialism, masculinity, and the heroic domination of nature—"not defeated but not in denial either about the level of destruction that we inherit and hold in our hand," whether we ask for it or not.^[11] In a scene from the film *Donna Haraway: Story Telling for Earthly Survival*, a playful exploration of her life and ideas, Haraway picks up a basket made by weavers from a Navajo tribe. As she handles the basket, she explains that she also touches and interacts with the tangled histories of colonial conquest, exploitation, and violence that characterize the United States' past and present.^[12] Her existence at that specific moment and place in time—as a white woman of Irish decent in Northern California in the twenty-first century—and her ability to handle, let alone possess that artifact, cannot be disentangled from these histories. We inherit, as she puts it, "the whole thing," whether we want to or not. Which inheritances, which externalities we choose to accept and allow to matter in narratives and representations becomes a political choice. "Storying otherwise," in Haraway's expression, is a characterization of the experiments with different kinds of storytelling, narratives that refuse the promissory narratives of progress to grapple with the destructive forces of settler-colonialism

and extractivism—the unfathomable violence of loss and grief and the relentless work of building livable worlds with what remains.

[10] Donna J. Haraway, *Staying with the Trouble: Making Kin in the Chthulucene* (Durham, NC: Duke University Press, 2016), 12.

[11] Lesley Instone and Affrica Taylor, "Thinking About Inheritance Through the Figure of the Anthropocene, from the Antipodes and in the Presence of Others," *Environmental Humanities* 7, no. 1 (2016): 133–50.

[12] Fabrizio Terranova, *Donna Haraway: Story Telling for Earthly Survival* (Brussels: Icarus Films, 2016).



NASA Apollo 17, Earth, 1972

From within the belly of environmental histories, speculative narratives assemble worlds with the matter of externalities by bending the documentary genre—the measured records of environmental documentation and climate communication—with the playfully surreal. Speculative fabulation is done with facts, many of which are stranger than fiction. Fabulation compresses facts—“composts” them, in Haraway’s expression—and does so in ways that produce, in the thickness of contradiction, accounts that allow for the exploration of irreverence and humor.

Like most stories, fabulations find truth through the imagination; they describe or explicate a here and now that is difficult to tell and hear to come to imagine worlds that are radically different than the ones we inhabit today, or create to echo the artist Robert Smithson, “true fiction that eradicates the false reality.”^[13]

This trajectory of “earth-writing”—geo (earth) and *graphia* (writing or drawing)—brings forth a renewed agency in the ways design culturally grapples with the climate crisis. Designing with externalities might be the apt characterization of this mode of “response-ability” for the worldmaking we participate in—bringing to matter those things, spaces, and lives in the geographic imagination.^[14] Approached as such, speculative narratives depict, in the presence of those people and places who have borne the consequences, how technological systems have changed the Earth to conjure *with* those externalities propositions for more inclusive worlds. The resulting Earth-drawing becomes a luring tale for a design agency to be sensitive to and to respond to who counts and what matters.

In this moment of climate crisis, design’s ability to respond to the world might hinge on a planetary imaginary that unsettles promissory environmental narratives and images—whether the histories of the Industrial Revolution and frontier settlement as well as future climate engineering. We might want to revisit, for example, one of the most widely circulated images of the Earth. In the Space Age, the iconic Apollo photographs—“The Whole Earth,” “Earthrise,” and “22727”—became a symbol of Earth as our “precious, fragile home,” presumably facilitating a vital environmental movement that animated popular desires to care for it.^[15] How might the whole Earth hold

[13] Robert Smithson, *Slideworks* (Verona: Carlo Frua, 1997), 80.
 [14] Rania Ghosn and El Hadi Jazairy, *Geostories: Another Architecture for the Environment* (New York: Actar, 2018).
 [15] In 1970, the first Earth Day was the largest single-day public protest to date in US history, with an estimated twenty million participants. The corresponding Earth Flag was inspired by the first photographs of the whole Earth taken during NASA’s Apollo 10 space mission in 1969. A later version of the Earth Flag used The Blue Marble, a NASA photograph that was taken by the astronauts of Apollo 17.
 [16] “Planet Earth by Dawn’s Early Light,” *Life*, vol. 61, no. 6, August 5, 1966.

its full promise of including everything and everyone without exception, a whole that truly implies that no thing has been omitted, abated, ignored, or taken away?

For another view of the planet, let’s consider a photograph from a series taken by the 1966 Earth-orbiting Gemini 10 shuttle. The photograph shows a single trash bag floating in space—a bag that contained the objects that NASA had intended to leave behind before the mission’s return to Earth.^[16] At over a million feet above the planet’s surface, the plastic bag seemed categorically external to concerns on Earth, as if it could safely disappear forever. Yet the material afterlives of space debris continue to haunt the Earth with an ever-increasing number of decommissioned satellites, extraterrestrial resource-mining missions, and other space debris. Today, scientists in space agencies consider that this space junk poses a threat not only to other space-based projects but also to people and property on Earth.

How might a planetary imagination, rather than seeking to forever keep that bag at bay, stage it within the portrait of the Earth? Using the language of science fiction, DESIGN EARTH engages the lives of technological objects, such as the management of space junk. In the “Neck of the Moon” project, you encounter Laika, a new satellite planet—named for the stray dog launched into low orbit by the USSR in 1957—made entirely of compacted space junk that orbits the Earth. Equipped with a giant robotic tug that approaches and compresses large pieces of debris at high altitudes, the compacted mass grows steadily and continuously, becoming a kind of second moon. In the “Cosmorama” project, the “Pacific Cemetery” triptych visualizes how decommissioned satellites and other space debris

are brought back from orbit at Point Nemo, a spacecraft cemetery known formally as the South Pacific Ocean Uninhabited Area, where it is terraformed to house refugees displaced from their nearby islands by the impacts of climate change. In response to the many “I can’t breathe” calls, the Respirator Goddess blows a breath of life into the planet.

To the question then of what the work of designers might be, one answer is that we need to make externalities matter, to arouse a different sensibility of the critical climate troubles, so to begin to design and inhabit the Earth with care and responsibility.



DESIGN EARTH, Planet after Geoengineering (Dust Cloud), 2021